

program that uses the scene manager will be made up of the following files:

Main program The small main program (that you write) must first create an instance of every scene identified in your program, then create an instance of the scene manager, passing a list of the scenes and a frame rate. To start your program, you call the `run()` method of the scene manager. For each new project you build, you must write a new main program.

Scene manager The scene manager is written for you and is available as the `SceneMgr` class in the *pyghelpers.py* file. It keeps track of all the different scenes, remembers which one is current, calls methods in the current scene, allows for switching between scenes, and handles communication between scenes.

Scenes Your program can have as many scenes as you want or need. Each scene is typically developed as a separate Python file. Each scene class must inherit from the prewritten `Scene` base class and have a set of methods whose names are predefined. The scene manager uses polymorphism to call these methods in the current scene. I have provided a template *ExampleScene.py* file to show you how to build a scene.

The code for the `SceneMgr` class and the code for the `Scene` base class live in the *pyghelpers* package. The scene manager is an object manager object that manages any number of `Scene` objects.

A Demo Program Using a Scene Manager

As a demonstration, we'll build a Scene Demo program that contains three simple scenes: Scene A, Scene B, and Scene C. The idea is that from any scene, you can click a button to get to any other scene. Figures 15-5 through 15-7 show screenshots of the three scenes.